Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy only when the stated facts allow it.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pendulum” and “potential energy” to explain your answer. Sentence starters are optional.

pendulum: an object that swings about a fixed support

potential energy: energy associated with the configuration of interacting objects

kinetic energy: energy associated with motion

01 Warm-up: bring back an earlier idea

Gravitational potential energy changes from **-15 J** to **-3 J**. Find ΔU and state whether energy is gained or lost.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: mg Δh = ½mv²; v = √(**2 g** Δh).
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest **0.8 m** above its lowest point reaches speed √(**2** gh)=**4 m/s** there with g=**10** and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial **Transfers** **Final**

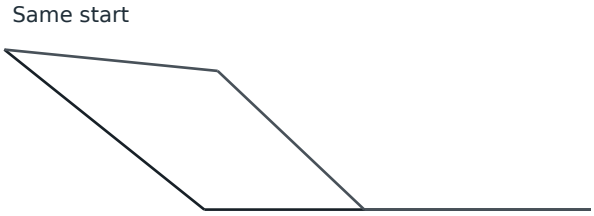
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two frictionless tracks start and end at the same heights but take different paths. **Compare final speeds** from rest.

Sentence starter: Both ... . They differ in ... because ... .



Mark the vertical height change.

Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart drops **3 m**.
1. Use g=**10**.
2. **Find lost potential energy** and gained kinetic energy.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A classmate uses total ramp length as h in mgh. **Explain how** to **name the correct measurement**.

Sentence starter: I think … . I can point to …, which shows …, so … .

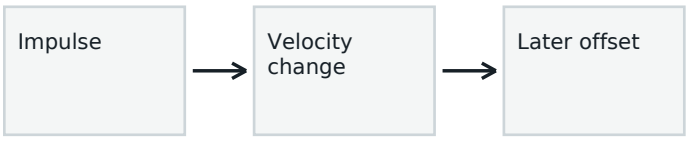
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A fixed impulse changes a target's sideways velocity by **0.04 m/s**. **Find the predicted shift after 600 s** and **state a limitation**.
Label the quantities in the diagram and say what it treats as true.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: 1 2 3

07 Exit check

A pendulum loses **12 J** of gravitational energy while moving down. Neglect losses. What is its kinetic energy gain? **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **1 kg** pendulum bob is released from rest **0.8 m** above the bottom, but reaches only **0.6 m** on the opposite side. Use g = **10 N/kg**. **Complete the turning-point energy account** and **find the decrease** in mechanical energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3